

Recursively differentiable quasigroups of orders 14 and 18: the last open cases of the Couselo–González–Markov–Nechaev conjecture

Alejandro Lizardi*

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Abstract

A quasigroup $(\Omega, \cdot)$ is *recursively differentiable* if its first recursive derivative $x * y := y \cdot (x \cdot y)$ is again a quasigroup. In 1998 Couselo, González, Markov, and Nechaev conjectured that a recursively differentiable quasigroup of order q exists for every $q \notin \{2, 6\}$, and verified this for all q except $q \in \{14, 18, 26, 42\}$. The case $q = 42$ was settled in 2008 and the case $q = 26$ in 2026, leaving $q = 14$ and $q = 18$ as the only open orders. We settle both by explicit construction. We exhibit a recursively differentiable quasigroup of order 14 and one of order 18, each invariant under a fixed-point-free automorphism with two regular orbits (a pair of 7-cycles and a pair of 9-cycles, respectively). Both are found by a prescribed-automorphism search and certified by the $O(n^2)$ test that defines the property; the full Cayley tables and their recursive derivatives are printed in the paper and re-checked by a standalone verifier. Together with the known cases this completes the conjecture: a recursively differentiable quasigroup of order q exists for every $q \notin \{2, 6\}$, and for no other order is the question open. The two orders resisted the standard perfect-Mendelsohn-design constructions for an arithmetic reason we make explicit (for those design constructions, $14, 18 \not\equiv 0, 1 \pmod{4}$ and $\not\equiv 0, 1 \pmod{5}$ — a gap in the *known methods*, not an obstruction to existence), which is why an ad-hoc, computer-assisted construction is the natural route to them.

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1 Introduction

A *quasigroup* of order n is a finite set Ω with a binary operation $\cdot$ whose Cayley table is a Latin square: every row and every column is a permutation of Ω . Equivalently, both equations $a \cdot x = b$ and $y \cdot a = b$ have unique solutions for all a, b .

*Independent researcher, alej@lizardi05@gmail.com. The constructions and computations in this paper were AI-assisted by Claude, under the Periapsis research structure. Both main results are explicit finite objects whose defining property is verifiable in $O(n^2)$ arithmetic; a self-contained verifier is provided with the ancillary files (Section 7).

Given a quasigroup $(\Omega, \cdot)$, Couselo, González, Markov, and Nechaev [1] defined a sequence of *recursive derivatives* $(\Omega, *_n)$ by

$$a *_{-2} b = a, \quad a *_{-1} b = b, \quad a *_n b = (a *_n b) \cdot (a *_n b) \quad (n \geq 0). \quad (1)$$

Thus $a *_0 b = a \cdot b$ is the original operation, and the *first recursive derivative* is

$$a *_1 b = (a *_{-1} b) \cdot (a *_0 b) = b \cdot (a \cdot b). \quad (2)$$

A quasigroup is *recursively n -differentiable* if $(\Omega, *_k)$ is a quasigroup for every $0 \leq k \leq n$; a recursively 1-differentiable quasigroup is called simply *recursively differentiable*. In the loop-and-quasigroup open-problems list [4] this object appears under the name *quasigroup with a quasigroup core*: the question, proposed by Syrbu at Loops '03, asks for which small orders a quasigroup Q exists such that the operation $x * y$ defined by (2) is itself a quasigroup operation. (The list writes the operation as $x * y = y - xy$, where the dash denotes the quasigroup multiplication, not subtraction; a general quasigroup has no inverses, so $\cdot$, not $^{-1}$, is the only operation available.)

By a theorem of [1], recursive $(n-3)$ -differentiability of a quasigroup of order q is equivalent to the existence of a full recursive $[n, 2, n-1]_q$ MDS code; the case $n = 4$, i.e. recursive differentiability, corresponds to a recursive $[4, 2, 3]_q$ code. Constructing an ordinary $[4, 2, 3]_q$ code amounts to a pair of orthogonal Latin squares of order q , which exist for all $q \notin \{2, 6\}$ (Bose–Shrikhande–Parker); the *recursive* constraint is what makes the differentiable version hard.

Conjecture 1.1 (Couselo–González–Markov–Nechaev [1]). *For every $q \notin \{2, 6\}$ there exists a recursively differentiable quasigroup of order q .*

The originators verified Conjecture 1.1 for all q except $q \in \{14, 18, 26, 42\}$. Markov, Nechaev, Skazhenik, and Tveritinov [2] settled $q = 42$ in 2008 with an explicit construction, and Klimov [3] settled $q = 26$ in 2026 via a perfect cyclic Mendelsohn design. This left exactly two open orders, $q = 14$ and $q = 18$. Our main result removes them.

Theorem 1.2. *There exists a recursively differentiable quasigroup of order 14 and a recursively differentiable quasigroup of order 18. Explicit Cayley tables are given in Sections 5 and 6.*

Corollary 1.3. *Conjecture 1.1 holds: a recursively differentiable quasigroup of order q exists for every $q \notin \{2, 6\}$, and these are exactly the orders for which one exists.*

Proof. The orders $q \in \{2, 6\}$ admit no pair of orthogonal Latin squares, hence no recursive $[4, 2, 3]_q$ code and no recursively differentiable quasigroup. All orders $q \notin \{2, 6, 14, 18, 26, 42\}$ are [1]; $q = 42$ is [2]; $q = 26$ is [3]; and $q \in \{14, 18\}$ is Theorem 1.2. $\square$

The verification underlying Theorem 1.2 is entirely elementary: each claimed object is a finite Cayley table, and the recursive-differentiability property is checked in $O(n^2)$ arithmetic directly from the definition (Section 2). The two tables were produced by a prescribed-automorphism backtracking search (Section 4); the search is how the objects were *found*, but it is not part of their *certification* — a reader may discard the search entirely and check the printed tables by hand or in a few lines of code.

Why 14 and 18 were the residue. Section 3 records the elementary arithmetic reason the standard constructions miss exactly these two orders, which both explains why they remained open and motivates the ad-hoc route we take.

2 The recursive derivative as a table operation

Throughout we identify a quasigroup of order n with its Cayley table $L: \Omega \times \Omega \rightarrow \Omega$, $\Omega = \{0, 1, \dots, n-1\}$, where $L[a][b] = a \cdot b$; L is a Latin square. From (2), the first recursive derivative has Cayley table

$$D[a][b] = a *_1 b = b \cdot (a \cdot b) = L[b][L[a][b]]. \quad (3)$$

Thus the derivative is obtained from L by a single nonlinear substitution: in row a , column b , look up the product $v = L[a][b]$, then read off $L[b][v]$. A quasigroup $(\Omega, \cdot)$ is recursively differentiable precisely when both L and the derived table D of (3) are Latin squares. This is the whole certificate, and it is checkable in $O(n^2)$ time.

Remark 2.1. The cyclic group $\mathbb{Z}_n$ has $L[a][b] = a + b$, so $D[a][b] = b + (a + b) = a + 2b$. As a map of b for fixed a this is a bijection iff $\gcd(2, n) = 1$, i.e. iff n is odd. Hence $\mathbb{Z}_n$ (and more generally any quasigroup that is “linear” over an abelian group with the doubling map appearing) is recursively differentiable exactly for odd n , and is *not* for even n . In particular the two orders we treat, being even, admit no such linear solution: a genuinely nonlinear table is required.

3 Why orders 14 and 18 resisted the standard constructions

The most productive known route to recursively differentiable quasigroups, used by Klimov [3] for $q = 26$, is the *directed standard construction* from a perfect cyclic Mendelsohn design. A $(v, k, 1)$ -PMD yields, by reading off consecutive triples of each cyclically ordered block, a recursively $(k - 3)$ -differentiable quasigroup of order v ; in particular a $(v, 4, 1)$ -PMD gives a recursively differentiable quasigroup of order v . The relevant existence facts are classical: a $(v, 4, 1)$ -PMD exists for all $v \equiv 0, 1 \pmod{4}$ (except $v \in \{4, 8\}$), and a $(v, 5, 1)$ -PMD for all $v \equiv 0, 1 \pmod{5}$ (except $v \in \{6, 10, 15, 20\}$); see the survey in [3] and the references therein.

This is exactly where 14 and 18 fall through:

$$14 \equiv 2 \pmod{4}, \quad 18 \equiv 2 \pmod{4}, \quad 14 \equiv 4 \pmod{5}, \quad 18 \equiv 3 \pmod{5},$$

so neither admits a $(v, 4, 1)$ - or a $(v, 5, 1)$ -PMD, and the block-size-4 and -5 PMD routes are simply unavailable. (By contrast $26 \equiv 1 \pmod{5}$, which is precisely what let [3] build a $(26, 5, 1)$ -PMD.) The other published families also miss the two orders: linear-recursive-sequence constructions live over prime-power alphabets and $14 = 2 \cdot 7$, $18 = 2 \cdot 3^2$ are not prime powers; the pseudogeometry method of [2] produced only $q = 42$. Klimov [3] accordingly lists 14 and 18 as the remaining open cases, to be approached “by other classes of methods.” The prescribed-automorphism search of the next section is such a method.

We stress that this is a statement about *known constructions*, not an obstruction to existence: most even orders *do* carry recursively differentiable quasigroups (e.g. 10, 12, 20, 22, 24, $\dots$ are all known), so there is no parity or congruence barrier in principle — 14 and 18 are merely the awkward composite orders that the congruence-restricted design constructions happen to skip. Theorem 1.2 confirms that existence holds for them too.

4 The construction method

Both objects were found by a backtracking search for a Latin square L such that the derived table D of (3) is also Latin, constrained to be invariant under a prescribed automorphism. A

permutation g of Ω is an *automorphism* of $(\Omega, \cdot)$ if $L[g(a)][g(b)] = g(L[a][b])$ for all a, b ; such a g partitions the n^2 cells of L into orbits under $(a, b) \mapsto (g(a), g(b))$, and fixing the value of one cell forces the values on its whole orbit. Searching only over orbit representatives shrinks the space by a factor of $\text{ord}(g)$ and tends to produce highly structured solutions.

We take g to be a fixed-point-free permutation with two regular orbits: for $n = 14$,

$$g_{14} = (0\ 1\ 2\ 3\ 4\ 5\ 6)(7\ 8\ 9\ 10\ 11\ 12\ 13),$$

two 7-cycles; for $n = 18$,

$$g_{18} = (0\ 1\ 2\ 3\ 4\ 5\ 6\ 7\ 8)(9\ 10\ 11\ 12\ 13\ 14\ 15\ 16\ 17),$$

two 9-cycles. (The choice mirrors the factorization $n = 2m$ with m the odd part: $14 = 2 \cdot 7$, $18 = 2 \cdot 9$.) The search assigns cells in a most-constrained-first order, propagates each assignment across its g -orbit and through the partial derivative table D (a conflict in either L or D prunes the branch immediately), and uses randomized restarts. The order-14 table was found in seconds; the order-18 table required randomized restarts and about half an hour of search on a laptop. Implementation, parameters, and run logs are in the ancillary files (Section 7); none of it is needed to verify the results.

5 A recursively differentiable quasigroup of order 14

Theorem 5.1. *The Cayley table L_{14} below is a quasigroup of order 14 whose first recursive derivative D_{14} is also a quasigroup. Hence a recursively differentiable quasigroup of order 14 exists.*

The base table L_{14} (rows indexed by the left factor $a = 0, \dots, 13$ top to bottom, columns by the right factor $b = 0, \dots, 13$ left to right):

0	2	7	5	10	12	9	6	4	13	3	1	11	8
10	1	3	8	6	11	13	9	0	5	7	4	2	12
7	11	2	4	9	0	12	13	10	1	6	8	5	3
13	8	12	3	5	10	1	4	7	11	2	0	9	6
2	7	9	13	4	6	11	0	5	8	12	3	1	10
12	3	8	10	7	5	0	11	1	6	9	13	4	2
1	13	4	9	11	8	6	3	12	2	0	10	7	5
8	4	13	0	2	7	5	10	3	9	1	12	6	11
6	9	5	7	1	3	8	12	11	4	10	2	13	0
9	0	10	6	8	2	4	1	13	12	5	11	3	7
5	10	1	11	0	9	3	8	2	7	13	6	12	4
4	6	11	2	12	1	10	5	9	3	8	7	0	13
11	5	0	12	3	13	2	7	6	10	4	9	8	1
3	12	6	1	13	4	7	2	8	0	11	5	10	9

Its first recursive derivative $D_{14}[a][b] = L_{14}[b][L_{14}[a][b]]$:

0	3	13	10	12	4	2	5	1	7	11	6	9	8
3	1	4	7	11	13	5	9	6	2	8	12	0	10
6	4	2	5	8	12	7	11	10	0	3	9	13	1
8	0	5	3	6	9	13	2	12	11	1	4	10	7
7	9	1	6	4	0	10	8	3	13	12	2	5	11
11	8	10	2	0	5	1	12	9	4	7	13	3	6
2	12	9	11	3	1	6	0	13	10	5	8	7	4
4	6	3	13	9	11	8	1	7	12	10	0	2	5
9	5	0	4	7	10	12	6	2	8	13	11	1	3
13	10	6	1	5	8	11	4	0	3	9	7	12	2
12	7	11	0	2	6	9	3	5	1	4	10	8	13
10	13	8	12	1	3	0	7	4	6	2	5	11	9
1	11	7	9	13	2	4	10	8	5	0	3	6	12
5	2	12	8	10	7	3	13	11	9	6	1	4	0

Proof. One checks directly that every row and every column of L_{14} is a permutation of $\{0, \dots, 13\}$, so L_{14} is a Latin square and $(\{0, \dots, 13\}, \cdot)$ with $a \cdot b = L_{14}[a][b]$ is a quasigroup. Computing $D_{14}[a][b] = L_{14}[b][L_{14}[a][b]]$ cell by cell yields the second table above (e.g. $D_{14}[0][2] = L_{14}[2][L_{14}[0][2]] = L_{14}[2][7] = 13$), and every row and column of D_{14} is likewise a permutation of $\{0, \dots, 13\}$. Hence the first recursive derivative is also a quasigroup, and L_{14} is recursively differentiable. Both Latin-square checks are finite and are reproduced by the verifier of Section 7. $\square$

Remark 5.2. L_{14} is noncommutative, has no two-sided identity (it is not a loop), and is nonassociative (not a group); its derivative differs from the base table. It is idempotent exactly on its first orbit (the diagonal reads $0, 1, \dots, 6$ on rows 0–6 and is fixed-point-free on rows 7–13), and is invariant under $g_{14} = (0 \cdots 6)(7 \cdots 13)$. Among the four readings of the “core” one might extract from a quasigroup — $b \cdot (a \cdot b)$, $a \cdot (a \cdot b)$, $(a \cdot b) \cdot b$, $(a \cdot b) \cdot a$ — only the correct one, $b \cdot (a \cdot b)$ of (2), produces a Latin derivative for this table; the object is keyed to the precise definition, not to a weaker variant.

6 A recursively differentiable quasigroup of order 18

Theorem 6.1. *The Cayley table L_{18} below is a quasigroup of order 18 whose first recursive derivative D_{18} is also a quasigroup. Hence a recursively differentiable quasigroup of order 18 exists.*

The base table L_{18} ($a, b = 0, \dots, 17$):

11	9	15	5	2	8	7	3	14	16	4	12	0	17	6	1	10	13
15	12	10	16	6	3	0	8	4	14	17	5	13	1	9	7	2	11
5	16	13	11	17	7	4	1	0	12	15	9	6	14	2	10	8	3
1	6	17	14	12	9	8	5	2	4	13	16	10	7	15	3	11	0
3	2	7	9	15	13	10	0	6	1	5	14	17	11	8	16	4	12
7	4	3	8	10	16	14	11	1	13	2	6	15	9	12	0	17	5
2	8	5	4	0	11	17	15	12	6	14	3	7	16	10	13	1	9
13	3	0	6	5	1	12	9	16	10	7	15	4	8	17	11	14	2
17	14	4	1	7	6	2	13	10	3	11	8	16	5	0	9	12	15
6	0	14	12	11	15	1	7	13	8	9	17	5	4	3	2	16	10
14	7	1	15	13	12	16	2	8	11	0	10	9	6	5	4	3	17
0	15	8	2	16	14	13	17	3	9	12	1	11	10	7	6	5	4
4	1	16	0	3	17	15	14	9	5	10	13	2	12	11	8	7	6
10	5	2	17	1	4	9	16	15	7	6	11	14	3	13	12	0	8
16	11	6	3	9	2	5	10	17	0	8	7	12	15	4	14	13	1
9	17	12	7	4	10	3	6	11	2	1	0	8	13	16	5	15	14
12	10	9	13	8	5	11	4	7	15	3	2	1	0	14	17	6	16
8	13	11	10	14	0	6	12	5	17	16	4	3	2	1	15	9	7

Its first recursive derivative $D_{18}[a][b] = L_{18}[b][L_{18}[a][b]]$:

12	14	10	9	7	1	15	6	0	16	13	11	4	8	5	17	3	2
1	13	15	11	10	8	2	16	7	3	17	14	12	5	0	6	9	4
8	2	14	16	12	11	0	3	17	5	4	9	15	13	6	1	7	10
9	0	3	15	17	13	12	1	4	11	6	5	10	16	14	7	2	8
5	10	1	4	16	9	14	13	2	0	12	7	6	11	17	15	8	3
3	6	11	2	5	17	10	15	14	4	1	13	8	7	12	9	16	0
15	4	7	12	3	6	9	11	16	1	5	2	14	0	8	13	10	17
17	16	5	8	13	4	7	10	12	9	2	6	3	15	1	0	14	11
13	9	17	6	0	14	5	8	11	12	10	3	7	4	16	2	1	15
7	15	2	10	14	0	8	9	5	13	11	4	17	1	3	12	6	16
6	8	16	3	11	15	1	0	10	17	14	12	5	9	2	4	13	7
11	7	0	17	4	12	16	2	1	8	9	15	13	6	10	3	5	14
2	12	8	1	9	5	13	17	3	15	0	10	16	14	7	11	4	6
4	3	13	0	2	10	6	14	9	7	16	1	11	17	15	8	12	5
10	5	4	14	1	3	11	7	15	6	8	17	2	12	9	16	0	13
16	11	6	5	15	2	4	12	8	14	7	0	9	3	13	10	17	1
0	17	12	7	6	16	3	5	13	2	15	8	1	10	4	14	11	9
14	1	9	13	8	7	17	4	6	10	3	16	0	2	11	5	15	12

Proof. As in Theorem 5.1: every row and column of L_{18} is a permutation of $\{0, \dots, 17\}$, so L_{18} is a quasigroup; the table $D_{18}[a][b] = L_{18}[b][L_{18}[a][b]]$ computed cell by cell is the second table, and every one of its rows and columns is a permutation of $\{0, \dots, 17\}$, so the derivative is a quasigroup. Hence L_{18} is recursively differentiable. The verifier of Section 7 reproduces both checks. $\square$

Remark 6.2. L_{18} is again idempotent-free, noncommutative, has no identity, and is nonassociative; it is invariant under $g_{18} = (0 \cdots 8)(9 \cdots 17)$, and only the reading $b \cdot (a \cdot b)$ of the core yields a Latin derivative.

This, with Theorem 5.1, proves Theorem 1.2 and hence Corollary 1.3.

7 Computational verification

The two theorems are finite statements verified directly from the definitions. The ancillary directory contains a standalone verifier, `rdiff_verify.py` (Python standard library only, no external data), which:

1. reads each printed table L_{14} , L_{18} (also stored as JSON);
2. checks that L is a Latin square (every row and column a permutation);
3. forms the derivative two independent ways — by the closed form $D[a][b] = L[b][L[a][b]]$ of (3), and by unrolling the recurrence (1) step by step — confirms the two agree, and checks that D is a Latin square;
4. runs the “wrong-reading” control: of the four candidate core operations only $b \cdot (a \cdot b)$ yields a Latin derivative, confirming the object is certified against the exact definition.

The verifier shares no code with the search that produced the tables, and a reader can reconstruct it in a few lines. The complete list of checks (each labelled to the claim it supports) and the search implementation are in the ancillary files; see `code/README.md`. The entire verification runs in well under a second.

8 Discussion

Corollary 1.3 closes the existence question of Conjecture 1.1 for binary quasigroups: a recursively differentiable quasigroup of order q exists if and only if $q \notin \{2, 6\}$. The two orders treated here were the last open cases on the loop-and-quasigroup problem list [4] (equivalently, the last two orders without a known full recursive $[4, 2, 3]_q$ code).

Two natural questions remain, neither needed for the conjecture but both of interest. First, the present objects are *ad hoc*: they are explicit witnesses found by search, not instances of a uniform family, and they do not *explain* existence at 14 and 18 the way a Mendelsohn-design construction explains it elsewhere. A uniform construction covering the orders $\equiv 2 \pmod 4$ that the PMD route skips would subsume the search. Second, both witnesses happen to carry a fixed-point-free automorphism with two regular orbits (a pair of $\frac{n}{2}$ -cycles with $\frac{n}{2}$ odd); whether *every* order $q \notin \{2, 6\}$ admits a recursively differentiable quasigroup with such symmetry, and more generally what the higher recursive-derivative degrees $v_r(2, q)$ are for 14 and 18 (we obtain only degree 1), is open. We hope the explicit tables are useful as test data for these questions.

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Honest status (read before relying on this). This is an unrefereed preprint with no journal referee check. Its central claims are finite and self-certifying: each Cayley table is printed in full and its recursive-differentiability is an $O(n^2)$ check reproduced by the standalone verifier, so the results are checkable without trusting the author or the assistant. The *novelty* claim — that no explicit order-14 or order-18 recursively differentiable quasigroup had previously been published — rests on a literature survey that reached the controlling sources ([3] first-hand, with its order- ≤ 100 table recording degree 0 at 14 and 18; the Moldovan-school survey tables recording “unknown” at both) but did *not* include a first-hand reading of the original 1998 source [1], which is in Russian and was consulted in translation/secondary form. A reader with access should confirm that point before treating the result as a priority claim rather than a clean construction.

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